Agenda

Python and R (for data science and emerging technologies)

(each section = 1 Hour Theory, 2 Hours Practice)

Part 1: Introduction and Basic Programming

1. **Introduction (to Data Science)**
    - Overview of data science, Python, and R
    - Setting up Python (Anaconda, Jupyter Notebooks) and R (RStudio)
    - Practice: Environment setup and basic operations in Python and R
2. **Basics of Python Programming**
    - Python syntax, basic constructs, data structures
    - Practice: 5 Python exercises covering these topics
3. **Basics of R Programming**
    - R syntax, basic constructs, data structures
    - Practice: 5 R exercises covering these topics

Part 2: Data Manipulation and Visualization

1. **Data Manipulation and Cleaning**
    - Handling data in Python (Pandas) and R (dplyr, tidyr)
    - Practice: 5 exercises on data cleaning and manipulation in Python and R
2. **Data Visualization**
    - Creating plots in Python (Matplotlib, Seaborn) and R (ggplot2)
    - Practice: 5 exercises on data visualization in Python and R
3. **Web-viewer for Dash-boarding**
    a. Tutorial Streamlit for Python
    b. Shiny in R (practical exercise replication)

Part 3: Statistical Analysis and Modeling

1. **Statistical Analysis and Modeling**
    - Basic statistics, linear and logistic regression in Python and R
    - Practice: 5 exercises on statistical analysis in Python and R
2. **Advanced Topics in Data Science**
    - Introduction to machine learning with Python (Scikit-learn) and R (caret)
    - Practice: 5 exercises on machine learning models in Python and R
3. **Capstone Project**
    - Applying concepts learned to a real-world data science problem
    - Encouragement to use both Python and R

Planning

Chapter 1: Introduction to Programming and Emerging Technologies

Part 1: Introduction and Basic Programming

Chap 1: Introduction

Introduction (1/2)

Python and R Usage Statistics

Introduction (1/2)

Python and R Loves/Dreaded Statistics

Introduction (1/2)

Python and R Jobs Statistics

Introduction (1/2)

Python and R in DS Statistics

Introduction (1/2)

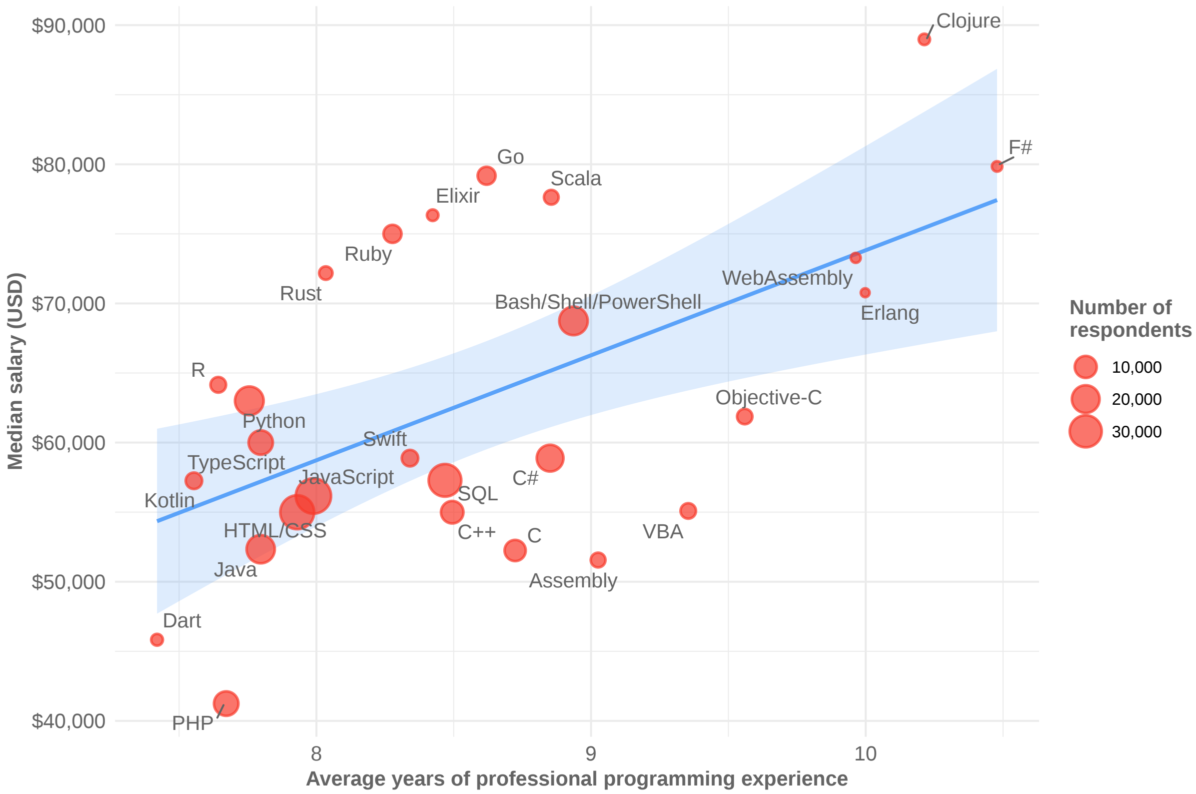
Python and R in DS Statistics

Introduction (1/2)

Python and R Salary Statistics

Introduction (1/2)

Python and R Salary Statistics (older)



Introduction (2/2)

Other stats

- Programming stats https://insights.stackoverflow.com/survey
- Data science stats: https://www.datacamp.com/blog/python-vs-r-for-data-science-whats-the-difference
- comparison https://s3.amazonaws.com/assets.datacamp.com/email/other/Python+vs+R.pdf

Setting up Python and R

- Anaconda (Python) https://www.anaconda.com/products/distribution
- Jupyter Notebooks: https://jupyter.org/install
- R: https://cran.r-project.org/
- RStudio: https://www.rstudio.com/products/rstudio/download/

Practice - Environment Setup

- Basic task: Type `1 + 2` in both Python and R environments.